Search Ad Attribution & Campaign Budget Allocation Analysis

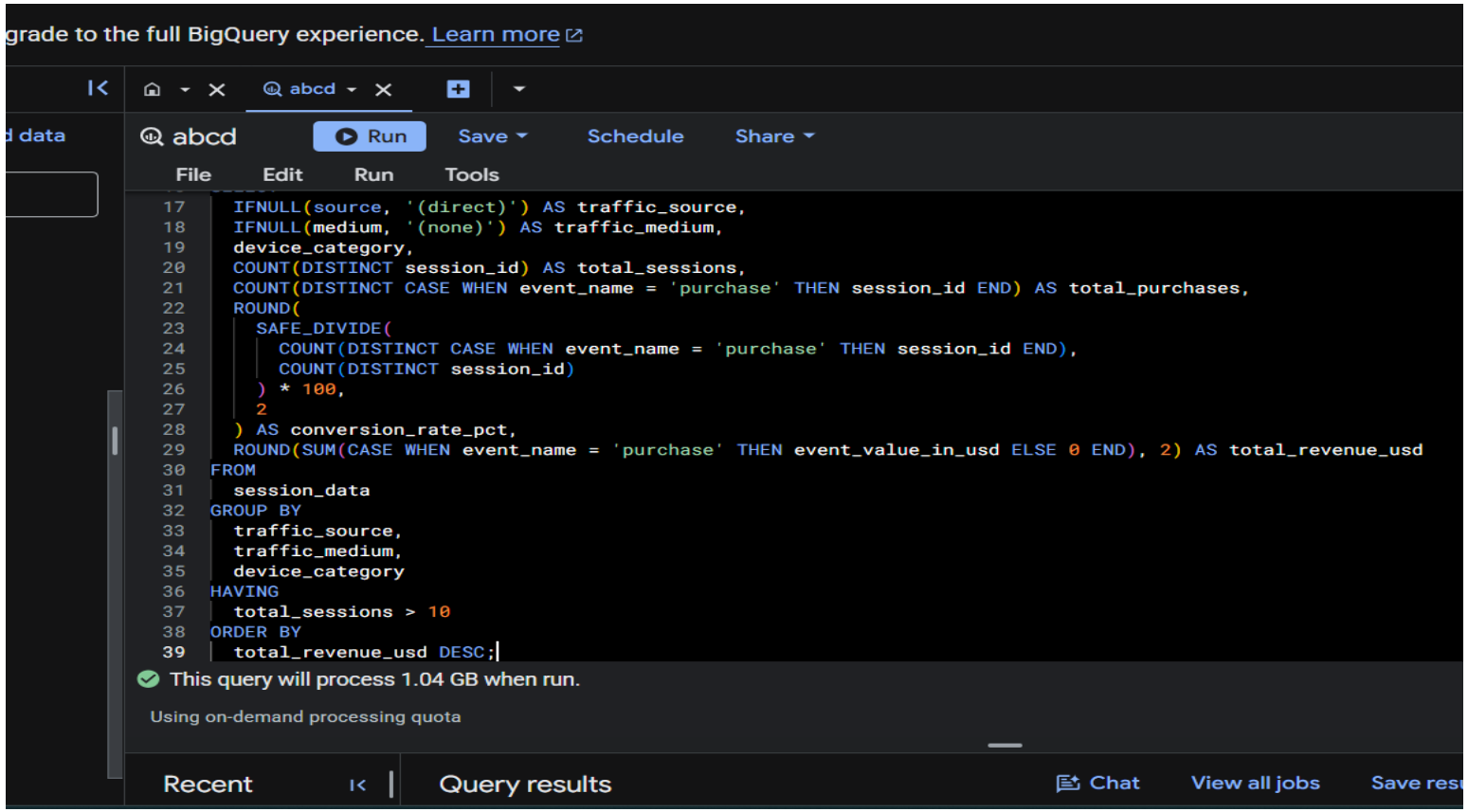
The Business Problem:

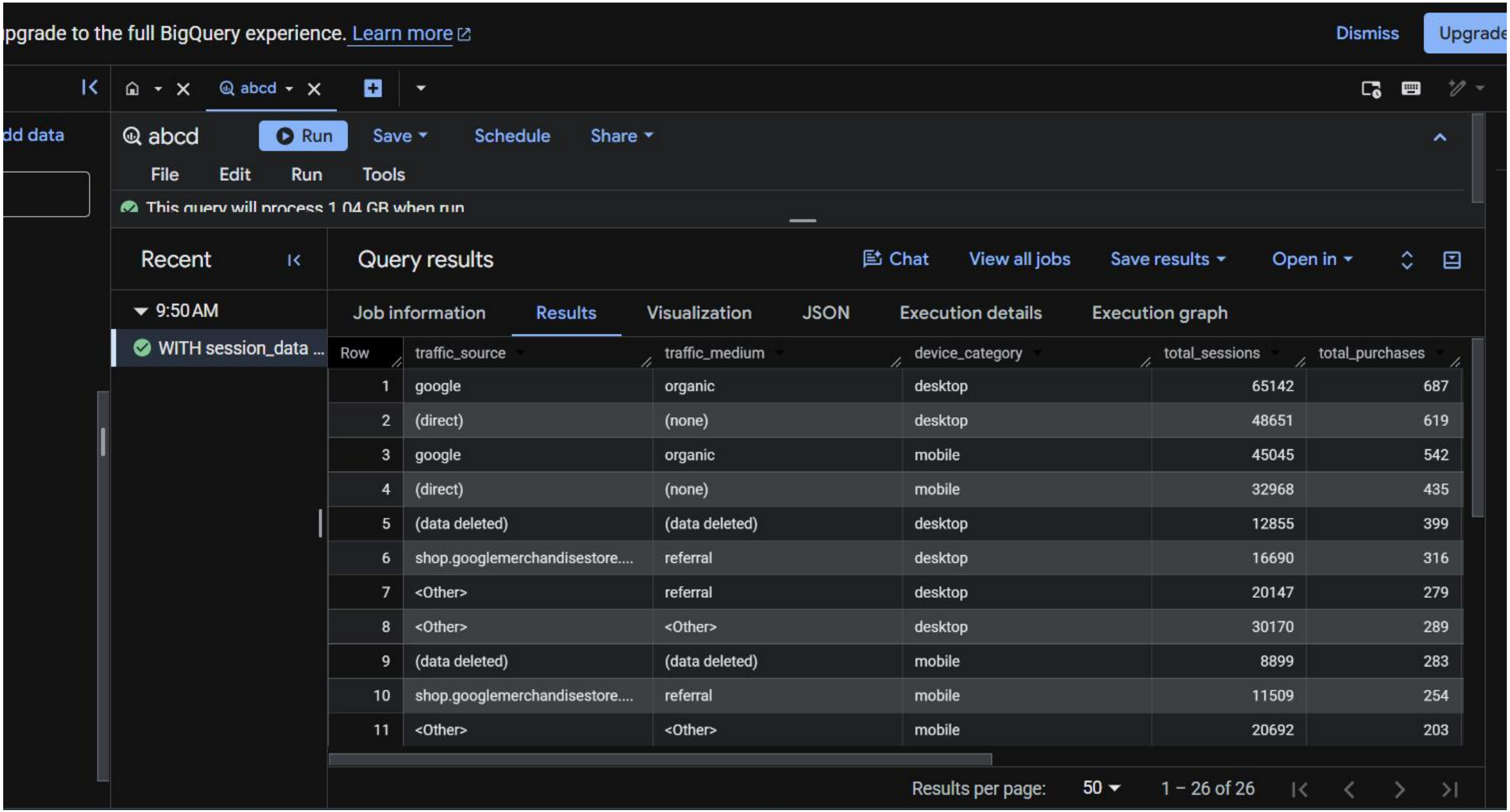
The marketing team spends budget across multiple Google Ads Search campaigns and acquisition channels, but leadership needs to determine which specific campaigns generate high-intent buyers versus low-value traffic, and where to reallocate budget to maximize ROAS.

The Strategy:

By analyzing full-funnel GA4 e-commerce data, marketing spend can be shifted toward high-converting traffic sources and away from underperforming channels, maximizing overall Return on Ad Spend (ROAS).

- This SQL query transforms raw, messy Google Analytics 4 (GA4) event data into a clean, easy-to-read business report that shows exactly which marketing channels are driving the most money.





Row	traffic_source	traffic_medium	device_category	total_sessions	total_purchases
1	google	organic	desktop	65142	687
2	(direct)	(none)	desktop	48651	619
3	google	organic	mobile	45045	542
4	(direct)	(none)	mobile	32968	435
5	(data deleted)	(data deleted)	desktop	12855	399
6	shop.googlemerchandisestore....	referral	desktop	16690	316
7	<Other>	referral	desktop	20147	279
8	<Other>	<Other>	desktop	30170	289
9	(data deleted)	(data deleted)	mobile	8899	283
10	shop.googlemerchandisestore....	referral	mobile	11509	254
11	<Other>	<Other>	mobile	20692	203

What the SQL code actually do:

- Extracted nested GA4 ga_session_id parameters from Google Cloud BigQuery raw datasets via UNNEST operations.
- Cleaned attribution parameters (traffic_source, traffic_medium, device_category) handling null values using IFNULL.
- Aggregated session counts, purchase events, and gross revenue using conditional aggregation and SAFE_DIVIDE.

Executive KPI & Campaign Attribution Dashboard(Data studio):

- Executive Scorecards: Deployed a real-time BI dashboard visualizing a $362K total revenue capture across 360K global e-commerce sessions. Campaign Attribution (Table): Mapped revenue against distinct traffic channels, identifying Google Search as the dominant paid acquisition engine generating $104K+ (29% of gross revenue).
- Platform Segmentation (Chart): Visualized severe revenue disparities across devices—desktop contributed $210K+, while mobile generated high session volume but under-monetized at $146K.
- Strategic Outcome: Delivered actionable device bid modifiers and channel reallocation data to immediately improve Return on Ad Spend (ROAS).

Data Studio Reporting - 11/7/26, 6:38 PM

File Edit View Insert Page Arrange Resource Help

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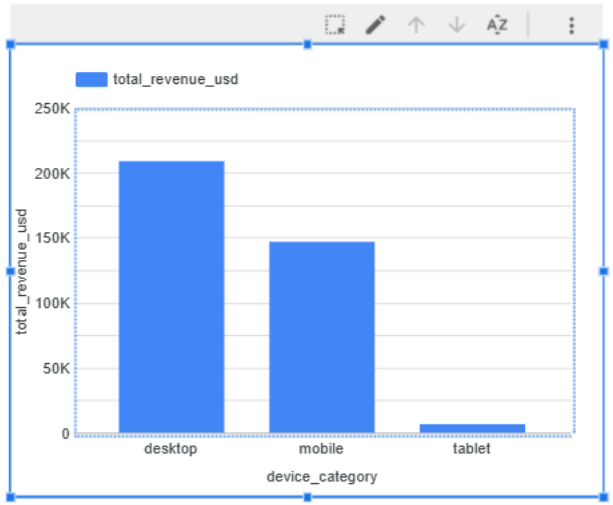
Add page | Add data | Blend | Add a chart | Add a control | Theme and layout

Add filter | Reset

total_sessions	total_revenue_usd	conversion_rate_pct
360,115	362,165	40.28

	traffic_source	total_revenue_usd	total_ses...
1.	google	$104,831	128,295
2.	<Other>	$81,099	97,279
3.	(direct)	$79,650	83,459
4.	(data deleted)	$50,064	22,233
5.	shop.googlemerchandisestore...	$46,521	28,849

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Bar chart properties

Search settings...

Chart types 49

Setup Style

(auto)

Show axis line

Grid

Grid background

Grid border color

Show X-axis gridlines

Show left Y-axis gridlines

Grid line style

Solid

Grid line color

Data

Search fields...

BigQuery Custo

conversion_rate

device_category

total_purchases

total_revenue_u

total_sessions

traffic_medium

traffic_source

Record Count

Add a field

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Add Data

- From the dashboard data, three key segment behaviors stand out:
- Desktop: Drives the overwhelming share of revenue (over $210K).
- Mobile: Accounts for substantial session volume (nearly 150K sessions) but yields only around $146K in revenue. This indicates a lower conversion efficiency or lower Average Order Value (AOV) on mobile despite high click spend.
- Tablet: Contributes negligible revenue (around $5K–$10K) despite consuming baseline budget across standard campaigns.

Source Attribution Disparity:

Google Search / Direct Channels: Strongest revenue drivers (Google alone accounts for $104,831; Direct accounts for $79,650).

<Other> and Referral Traffic (shop.googlemerchandisestore.com): Generates significant traffic and revenue ($81K and $46.5K respectively), but paid support here often overlaps with existing brand equity.

- Specific Budgetary Allocation Recommendations:

Action:

- Increase Google Ads bid adjustments on Desktop by +15% to +20%.
- Decrease Google Ads bid adjustments on Mobile by -15% to -25%.
- Decrease Tablet bid adjustments by -50% to -80%.

Reallocate 20% of Underperforming Spend to High-Intent Search

Action:

- Trim spend on broad match / top-of-funnel discovery campaigns that drive generic mobile traffic.
- Reallocate that capital directly into high-intent Google Search campaigns (exact/phrase match) focused on high-margin product categories.